

FAGI: Forecast Anywhere for Groundwater Interpolation

Summary as of March 03, 2026

blue = finished or running

orange = planned

Models

Component	Models	Train + validation metric	Final evaluation metric	HPO
Temporal (separate)	TFT, GRU	MSE	MAE, RMSE, nRMSE, NSE	Kunz et al. setup as reference so far, manual small grid search over key params (lr, layers, dropout, hiddensize) planned
Spatial (separate)	Kriging / GP	Kernel params fitted via GP marginal likelihood on training data	Same, plus Skill_vs_5NN_mean, Skill_vs_all_train_mean	Inner automatic local kernel optimization per fit; manual outer comparison planned (kernel type, isotropy, pretrain steps, pretrain lr, jitter)
Joint spatiotemporal (planned)	GRU+GP	Joint loss: GRU_MSE + gp_weight * GP_MSE	Same as spatial row	Manual tuning of gp_weight + selected GRU/GP learning params on validation

Splits

Split Type	Explanation
Temporal split	Fixed time-based split with Train <= 2016-01-01, Validation (2016-01-01, 2020-01-01], Test > 2020-01-01, based on Kunz et al.
Spatial split 1: Static-features cluster-based distributed (current split)	Cluster wells by static features and randomly assign a set fraction of wells from each cluster into train/holdout with fixed seed
Spatial split 2: Random split	Randomly assign wells to train/holdout with a set fraction and fixed seed
Spatial split 3: Grid-block split (harder)	Partition 2d space into grid blocks and assign fraction of blocks to train/holdout so generalization becomes harder (potentially add a buffer zone)
Spatial split 4: Spatially stratified + proximity constraint (easier)	Stratify spatially for coverage, then enforce maximum train–holdout distance to make generalization easier (since wells with very bad kriging performance seem to have bigger distances to closest training wells)

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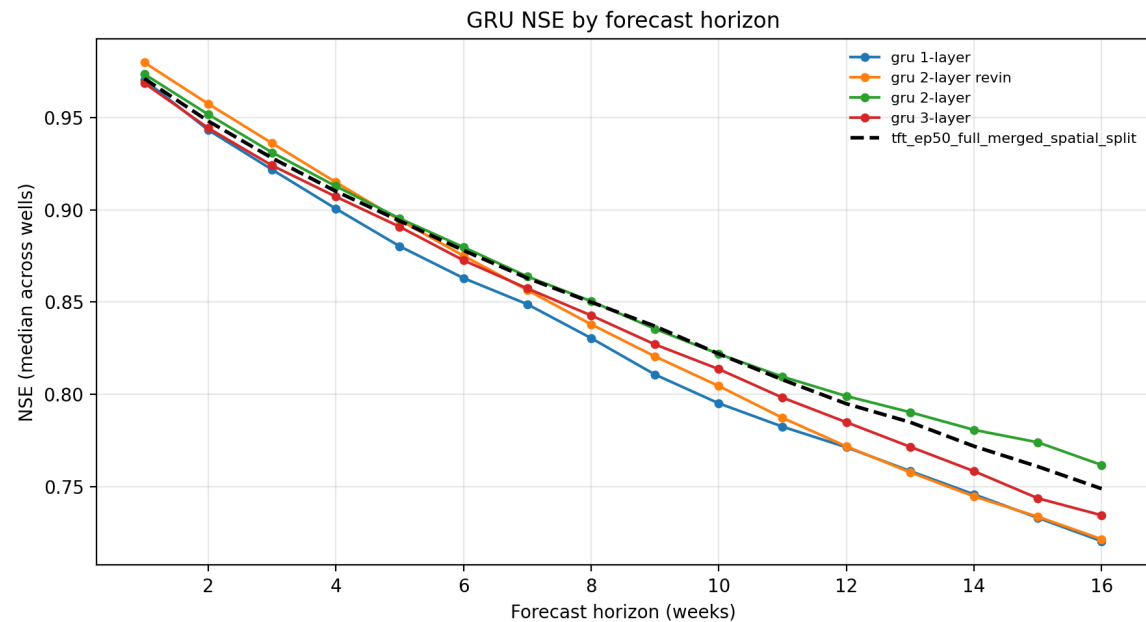
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Current Status:

- Successfully matched performance of GRU to TFT performance by closely matching TFT hyperparameters (see plot)
- No stable run for custom GP yet; currently running small HPO to find a stable config; will run bigger HPO after
- Introduction of lr scheduler and revin let performance for GRU drop (see plot); currently running ablation tests to find cause
- Small first HPO for GRU (max. 12 runs, see table) will run directly after
- Most current conclusions: GRU competitive with TFT; spatial performance sensitive to nearest train well distance

Hyperparameter	Values
model.gru_layers	1, 2, 3
model.gru_hidden	96, 128, 192
model.gru_dropout	0.1, 0.2, 0.3
training.lr	1e-4, 3e-4, 1e-3
max_trials	12
objective	RMSE @ horizon=16
fixed settings	in_len=52, out_len=16, batch_size=4096, seed=40



Next steps:

1. Proper HPO on GRU and GP
2. Finish implementation of joint training GRU+GP once both are stable and performing well, execute training and evaluate
3. Potentially HPO on joint training, depending if performance is already better than separate pipeline (otherwise postpone)
4. Implement missing 3 splits after comparison of separate and joint training on split 1
5. Find best configs for separate vs. joint training on all 4 splits
6. Conduct robust runs for all 8 configurations (meaningful number of seeds to be determined from previous runs)
7. Compare performances on holdout wells and find explanations for outliers/potentially reduce number of outliers